Class Meeting Handout 13

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Hammett plots are not always simple. In some cases, they can be curved. When this happens, we have an exciting and interesting observation to explain.[1]

This document was produced using the LaTeX typesetting language with the Tufte-handout document class. Chemical diagrams were created in *ChemDoodle*. Plots were typeset using *MatPlotLib* and *Python* tools. Diagrams were created and edited using *Affinity Designer*.

[1] Read chapter 8.3, 8.4 and 8.6.

LFER Part 3: Curved Plots

There are two ways a plot can be curved. If there are two different mechanisms for a reaction, and they have different reaction constants, we may see one dominate and then the other as we change substituents.[2] If there is a single mechanism with multiple steps, and each step has a different dependence on electron density, then we may see the rate-determining step change as we change substituents. The rate is limited by the slowest step, so we will see only the slower rate. Let us explore these ideas.

[2] This is similar to pH-rate profiles where multiple pathways are operating and we only see the largest contributor in each region of the plot. In a log-log plot, we are exploring rates with orders of magnitude in the difference.

Exercise: Curved Hammett Plots – 40 minutes

1. Create two linear Hammett plots that cross (have the same rate) when we are using m-Cl as the substituent. Plot them both on the same graph.[3]

   (a) Add the two plots together over a range of σ values. This is analogous to two different mechanisms contributing to the overall rate. What does it look like?

   (b) Only plot the line with the lesser rate of the two over a range of σ values. This is analogous to the slower reaction being the r.d.s. What does it look like?

[3] Recall your high-school math. How do you obtain a line equation for a line with a given slope that passes through a given point? Just do that twice.

2. Consider the hydrolysis of nitrosamines as shown in figure 1. These are carcinogenic byproducts of keeping red meat red by adding nitrite salts.[4] The reactions of such compounds in water would be important to study. Let us consider a Hammett plot of the hydrolysis of nitrosamines in acidic conditions. I found data for this reaction in a textbook[5] which referenced a very obscure Russian journal article as the source of the data.[6] I have not yet been able to find this article. I cannot even prove that this journal ever existed. So I digitized the data from Figure 4.5 in the textbook.

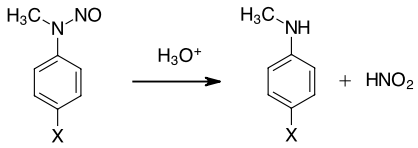
   (a) Propose a mechanism for the acid hydrolysis of a nitrosamine.

   (b) Now propose a different mechanism. For example, if the first mechanism features protonation followed by departure of the nitroso group, this one could have departure precede protonation.

   (c) Plot the data presented in table 2 on page 5 as a Hammett plot. What do you observe? Determine values for the reaction constant(s), ρ.

   (d) Reconcile your observation with the proposed mechanisms.

Figure 1: Acid hydrolysis of a nitrosamine. ↓



[4] "An Organic Chemist's Guide to N-Nitrosamines: Their Structure, Reactivity, and Role as Contaminants." J.C. Beard, T.M. Swager, *J. Org. Chem.*, **2021**, *86*, 2037-2057. https://doi.org/10.1021/acs.joc.0c02774

[5] "Physical Organic Chemistry, 2nd ed." by Neil Isaacs. **1995**, *Wiley, New York* (the 1st edition is available in UPEI library stacks QD476.I846 1987.)

[6] B.A. Porai-Koshits, E.Y. Belyaev, E. Shadovskii, *Reacts. Sposobnost. Org. Soedin.*, **1964**, *1*, 10.

3. We have seen the hydrolysis of benzyl chlorides and used Hammett plots to explore the mechanism. Let us now consider the solvolysis of benzyl tosylates in acetic acid[7] as shown in figure 2.

   (a) What is the tosylate group, and why is it a common group in synthetic organic chemistry?

   (b) Based on what you know about solvolysis of benzyl chloride in water and ethanol, propose a mechanism for this reaction.

   (c) Table 3 on page 5 presents rate data for a series of substituted benzyl tosylates. Plot this data as a Hammett plot. What do you observe?

   (d) The mechanism is believed to be S_N2 for benzyl tosylates in acetic acid because inversion is observed at the benzyl carbon. Given this fact, propose an explanation for the Hammett plot.[8]

4. The hydrolysis of imines is a great subject for physical organic chemistry. The rate data for the hydrolysis of a set of imines based on substituted anilines[9] is presented in table 4 on page 6. The reaction scheme is shown in figure 3.

   (a) Based on what you know about the formation and hydrolysis of imines in water, propose a mechanism for this reaction.

   (b) Plot the rate data from table 4 as a Hammett plot. What do you observe?

   (c) Explain how the proposed mechanism is consistent with the Hammett plot.[10]

5. Now let us examine the cyclodehydration of 2-phenyltriarylcarbinols[11] as shown in figure 4 on the following page. The rate data is collected in table 5 on page 6

   (a) Based on what you know about the formation of carbocations and electrophilic aromatic substitution, propose a mechanism for this reaction.

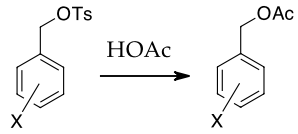
   (b) Plot the rate data from table 5 as a Hammett plot. What do you observe?

   (c) Explain how your proposed mechanism is consistent with the Hammett plot.[12]

That was a lot of plotting, and it is highly unlikely that we will be able to explore all four data sets described above during the class meeting. You should complete any exercises that we miss and compare your conclusions with your peers. You will learn much more about data analysis if you do it yourself.

[7] "Acetolysis reactivities of substituted benzyl and polycyclic arylmethyl p-toluenesulfonates. Correlations with SCF-π and CNDO MO methods." A. Streitwieser Jr., e. al., *J. Am. Chem. Soc.*, **1970**, *92*, 5141-5150 https://doi.org/10.1021/ja00720a025

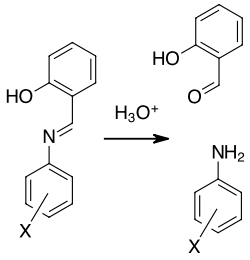
Figure 2: Acetolysis of benzyl tosylates. ↓



[8] Ch. 11.5 discusses S_N1/S_N2 substitutions. Ch. 11.5.8 has a nice summary.

[9] "Kinetics of hydrolysis of substituted salicylideneanilines." J. Hoffmann, J. Klicnar, V. Štěrba, M. Večeřa, *Coll. Czech. Chem. Comm.*, **1970**, *35*, 1387–1398. http://doi.org/10.1135/cccc19701387

Figure 3: Hydrolysis of salicylideneaniline.↓



[10] Ch. 10.15 discusses imine formation and catalysis.

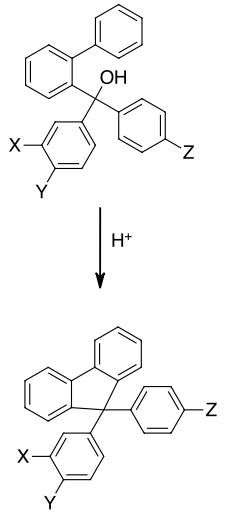
[11] "Mechanism of cyclodehydration of 2-phenyltriarylcarbinols", H. Hart, E.A. Sedor, *J. Am. Chem. Soc.*, **1967**, *89*, 2342-2347. https://doi.org/10.1021/ja00986a018

[12] Ch. 10.18 discusses electrophilic aromatic substitution and ch. 9.2 discusses the rate acceleration resulting from intramolecular mechanisms.

Discussion: Preparing for Your Exploration – 10 minutes

In class, we will discuss expectations for the exploration report and hopefully start you on your way to writing your own. Along with this handout, explorations of literature contributions featuring curved Hammett plots are available on the Moodle site. These are examples of the kind of work that I will be looking for when you submit your exploration report six weeks from now. We will revisit these explorations when we begin the second phase of this course, where we will apply the principles of physical organic chemistry to as many examples as possible. Any requests?

Figure 4: Cyclodehydration of 2-phenyltriarylcarbinols.↓



Learning Goals

Through the suggested textbook reading, problems, participating in this class discussion, and reading the sample exploration report, students will…

- have explored more complex reaction schemes and observed how LFER plots can curve depending on the changing relative rates of the steps or pathways.
- have gained more experience interpreting literature examples and following descriptions of chemical synthesis.
- **Computer Skill:** have again used an interactive *Python* notebook to interpret and analyze experimental data from the literature.

A Look Ahead

We have completed our investigation of the electronic effects of substitution. But, the whole Hammett plot idea is rigged by the use of only m- and p-substituted benzene rings as probes for electronic effects. In the Hammett approach, the steric demand of the phenyl group with respect to the reaction centre never changes. What if our changing groups also had very different steric interference? We will now need a score for steric effects.

In the next three meetings of lesson #3, we will survey other LFER systems and use parameters that describe the steric effects of substituents and the effects of solvents on the rates of a reaction. We will also enjoy a brief overview of the many other clever methods one could use to design an experiment for investigating the mechanism of a reaction.

Explorations to Come

The data sets presented today are from literature contributions. For the data sets presented today I have written the following exploration proposals[13] that are available via the moodle site.

When I go and find data, I often end up travelling down a rabbit hole as I try to interpret exactly how the data was obtained and what it means. I will follow references that the author claim support their conclusions and the rabbit hole gets deeper. Almost always, I find small errors or decide to interpret the data differently. Then I write a critical review of the work so that I have a record of my thoughts and analysis.

[13] Over time, I add to these documents and they are in various states of completion as fully fledged explorations. One or two might be good examples of an exploration as I would expect for your final project.

1. "Plots for Hydrolysis of Nitrosamines"

   I can't explore this work because I cannot get access to the original paper. I just stole the data from a secondary source.[5,6] This short exploration is just a presentation of the plots of that data.

2. "Exploration: Solvolysis of Benzyltosylates"

   In this exploration, I replotted the data and performed a curve fit optimization that I felt was more robust than the "drawing straight lines with a ruler" method that was used by the authors.[7] I revealed that the evidence of the curved line that supported a change in mechanism could be interpreted to be the deviation of a single point. This document is short and some more exploration of related work would be required to reach the level of effort expected for a student exploration report.

3. "Plots for Hydrolysis of salicylideneanilines"

   I have not explored this literature contribution[9] except to grab the data. I read some very recent work in which salicylideneanilines are being used as photoswitchable molecules. Also, the authors present data for the alkaline hydrolysis and plot pH-rate profiles. There are references to some other works that hint at some very interesting results from other authors. Perhaps this could be a thread that you pull on as you seek a topic for your own explorations.

4. "Exploration: Cyclodehydration"

   In this exploration of the work by Hart and Sedor[11] I plotted the data reported in the paper and compared it to the plot presented by the authors. I observed an error in the data table. I also re-analyzed the isotope effect data to strengthen my understanding of isotope effects. I then used the data presented by the authors to calculate the order of the reaction in acid activity. To do this, I had to find reported values for the acidity of acetic and sulfuric acid mixtures. All of my analyses were supportive of the author's conclusions.

Data Tables

The data for the exercises is collected in this section.

Substituent	σ	σ^+	σ^-
p-OCH_3	−0.27	−0.78	−0.26
m-OCH_3	0.11		
p-CH_3	−0.14	−0.31	−0.17
m-CH_3	−0.06		
H	0		
p-F	0.06	−0.07	−0.03
p-C_6H_5	0.06		
p-Cl	0.23	0.11	0.19
p-Br	0.23	0.15	0.25
m-Cl	0.37		
m-CF_3	0.46		
p-CF_3	0.53		0.65
p-CO_2Et	0.54		0.64
p-NO_2	0.78		1.24

Table 1: σ values for selected substituents. ⟵ The data in this table can be obtained from my GitHub repository at https://github.com/blinkletter/LFER-QSAR/blob/main/data/LFER_Williams.csv

Substituent	k /10^{-5} s^{-1}
p-OCH_3	11.59
p-CH_3	6.03
H	2.04
p-Ph	0.52
p-F	1.58
p-Cl	2.05
p-Br	2.38
p-COOEt	57.41
p-NO_2	293.76

Table 2: Acid hydrolysis of a nitrosamine in acidic water. Conditions unknown. Data is digitized from reference 6. ⟵ The data in this table can be obtained from my GitHub repository at https://github.com/blinkletter/4410PythonNotebooks/blob/main/Class_16/data/Nitrosamine.csv

Substituent	k /10^{-5} s^{-1}
p-CH_3	96.3
m-CH_3	4.54
p-F	4.19
H,	1.71
p-Cl	0.86
m-Cl	0.087
m-CF_3	0.0446
p-CF_3	0.0277

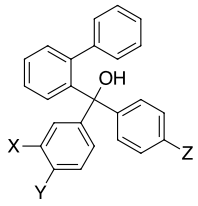
Table 3: Rates of solvolysis of benzyl toyslates in acetic acid at 40 °C. Data is from reference 7 ⟵ The data in this table can be obtained from my GitHub repository at https://github.com/blinkletter/4410PythonNotebooks/blob/main/Class_16/data/Table_1_data.csv

Substituent	k /10^{-3} min^{-1}
p-OCH_3	4.07
p-CH_3	7.76
m-CH_3	8.71
H	12.59
m-OCH_3	14.79
p-Cl	12.02
m-Cl	13.49
m-NO_2	7.08

Table 4: Rates of hydrolysis of salicylideneanilines at pH 5.5. Data is from reference 9
←
The data in this table can be obtained from my GitHub repository at https://github.com/blinkletter/4410PythonNotebooks/blob/main/Class_16/data/ImineHydrolysis.csv

Reactant	X	Y	Z	$\sum\sigma$	$\sum\sigma^+$	k_{rel}
1	H	p-Cl	p-Cl	0.46	0.22	0.096
2	m-CF_3	H	H	0.43	0.43	0.052
3	m-Cl	H	H	0.37	0.37	0.100
4	H	p-Cl	H	0.23	0.11	0.30
5	m-OCH_3	H	H	0.12	0.12	0.74
6	H	H	H	0	0	1.00
7	m-CH_3	H	H	-0.07	-0.07	1.48
8	H	p-CH_3	H	-0.17	-0.31	3.27
9	H	p-OCH_3	H	-0.27	-0.78	0.56
10	H	p-CH_3	p-CH_3	-0.34	-0.62	5.80
11	H	p-CH_3	p-OCH_3	-0.44	-1.09	0.176
12	H	p-OCH_3	p-OCH_3	-0.54	-1.56	0.009

Table 5: Rates of cyclodehydration of 2-Phenyltriarylcarbinols in 80 % aqueous acetic acid containing 4 % sulfuric acid. Rates are reported as relative rates. The rate constant for the unsubstituted case, **6**, is 2.54 × 10^{-3} s^{-1}. Data from reference 11 .
←



The data in this table can be obtained from my GitHub repository at https://github.com/blinkletter/4410PythonNotebooks/blob/main/Class_16/CyclodehydrationNotebooks/data/hammettdata.csv